

“Heart Disease Prediction using ML”

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled “**Heart Disease Prediction using ML**” is the bonafide work of “**Vedansh Sharma (23BCE10754), Preeyansh Arora (23BCE10813), Apan Sirohi (23BCE10832), Lakshya Nain (23BCE10833), Anirban (23BCE10856)**” who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Last, but not least, I am deeply indebted to my parents who have been the greatest support for the project to make it a success.

LIST OF ABBREVIATIONS

1. cp: chest pain type, 1: typical angina, 2: atypical angina, 3: non-anginal pain, 4: asymptomatic
2. trestbps: resting blood pressure
3. chol: serum cholesterol in mg/dl
4. fbs: fasting blood sugar > 120 mg/dl
5. restecg: resting electrocardiographic results (values 0,1,2)
6. thalach: maximum heart rate achieved
7. exang: exercise induced angina
8. oldpeak: oldpeak = ST depression induced by exercise relative to rest
9. slope: the slope of the peak exercise ST segment
10. ca: number of major vessels (0-3) colored by fluoroscopy
11. thal: thal: 3 = normal; 6 = fixed defect; 7 = reversible defect

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ABSTRACT

Heart disease remains one of the leading causes of mortality worldwide, making early detection and risk assessment crucial for timely intervention. Traditional methods of diagnosing heart disease often rely on subjective interpretation of medical data, which can be time-consuming and prone to human error. This paper presents a machine learning-based approach for heart disease detection, aimed at improving the accuracy and efficiency of early diagnosis. By leveraging patient data, such as age, blood pressure, cholesterol levels, and other vital health parameters, machine learning algorithms such as Random Forest, Support Vector Machines (SVM), and Logistic Regression were used to develop a predictive model for heart disease classification. The dataset used includes both categorical and numerical features, and preprocessing techniques like feature scaling and one-hot encoding were employed to prepare the data for analysis. The model's performance was evaluated using various metrics, including accuracy, precision, recall, and F1-score, with a focus on minimizing false negatives to ensure high sensitivity in detecting at-risk individuals. The results demonstrate that machine learning models can achieve high accuracy in predicting heart disease, providing healthcare professionals with an effective tool for early intervention. Furthermore, the system has potential applications in real-time monitoring and preventative healthcare, where individuals can assess their heart disease risk based on personal health data. The findings suggest that machine learning has the capability to transform healthcare diagnostics, making it more accessible, efficient, and precise.

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CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

Heart diseases are a leading cause of mortality worldwide. Therefore, it is critical to detect the diseases early and prevent them to reduce the mortality rate and improve the quality of life for individuals at risk. Machine learning (ML) offers powerful tools to analyze vast amounts of medical data and identify patterns that can help predict heart disease. By leveraging advanced algorithms, we can enhance diagnostic accuracy and support clinical decision-making processes.

1.2 Motivation for the Work

The motivation for this project stems from the need to address the rising prevalence of heart disease globally, improve early detection methods using technology, reduce the burden on healthcare systems by enabling automated and accurate predictions, utilize data-driven approaches to provide personalized insights for preventive care, bridge the gap between medical research and real-world application using ML techniques.

1.3 About the Project

This project explores the application of machine learning techniques in predicting heart disease. It involves collecting and preprocessing data from relevant sources, such as electronic health records or publicly available datasets, employing feature engineering to identify key predictors of heart disease, including age, cholesterol levels, blood pressure, and lifestyle factors, implementing various machine learning algorithms, such as logistic regression, decision trees, random forests, support vector machines (SVM), and deep learning techniques, evaluating the performance of these models using metrics like accuracy, precision, recall, F1-score and developing a user-friendly interface for clinicians or researchers to input patient data and receive predictions.

1.4 Problem Statement

Heart disease often remains undiagnosed until it reaches an advanced stage, leading to severe health consequences and even death. Traditional diagnostic methods are time-consuming, prone to human error, and may not provide a comprehensive risk assessment. Therefore, there is a pressing need for a reliable and automated system to predict heart disease at an early stage using data-driven approaches.

1.5 Objective of the Work

The objectives of this project are to compare machine learning-based models capable of predicting the likelihood of heart disease, identify the most significant factors contributing to heart disease risk, ensure the model's accuracy, interpretability, and ease of use in real-world scenarios and demonstrate the feasibility of integrating machine learning tools into clinical practice.

1.6 Organization of the Project

The project is organized into the following sections:

1. **Introduction:** Overview of the problem and the role of machine learning in solving it.
2. **Literature Review:** A summary of related work and existing methodologies.
3. **Methodology:** Details of the dataset, feature engineering, and ML algorithms used.
4. **Implementation:** Step-by-step process of model building and deployment.
5. **Results and Discussion:** Evaluation of model performance and insights derived.
6. **Conclusion and Future Work:** Summary of findings and potential future improvements.

1.7 Summary

This project aims to harness the potential of machine learning to predict heart disease effectively. By analyzing patient data and identifying key risk factors, the developed model seeks to enhance early diagnosis and improve patient outcomes. The successful completion of this work can significantly contribute to the field of predictive healthcare and inspire further advancements in ML-based medical applications.

CHAPTER 2: RELATED WORK INVESTIGATION

2.1 Introduction

The prediction of heart disease using machine learning has been an active area of research, with numerous studies exploring various techniques to enhance diagnostic accuracy. This chapter reviews related work in this domain, highlighting core areas of focus, existing approaches, and their respective advantages and limitations. By analyzing these methods, we aim to identify gaps and opportunities for improvement in current prediction systems.

2.2 Core Area of the Project

The core area of this project revolves around applying machine learning techniques to predict heart disease by analyzing patient data. This involves leveraging supervised learning models, feature selection methods, and performance evaluation metrics to build reliable and interpretable predictive systems. The key goals include improving the accuracy, scalability, and applicability of these models in clinical settings.

2.3 Existing Approaches/Methods

2.3.1 Approaches/Methods - 1: Logistic Regression

Logistic regression is one of the most commonly used algorithms for binary classification problems such as heart disease prediction. It estimates the probability of a patient having heart disease based on input features such as cholesterol levels, blood pressure, and lifestyle factors.

- Advantages: Simple, interpretable, and computationally efficient.
- Limitations: Assumes linear relationships between features and the target variable, which may not always hold true.

2.3.2 Approaches/Methods - 2: Decision Trees and Random Forests

Decision trees classify patients by splitting data based on feature thresholds, while random forests combine multiple decision trees to improve accuracy and reduce overfitting.

- Advantages: Handles non-linear relationships and provides feature importance rankings.
- Limitations: Susceptible to overfitting (decision trees) and high computational cost (random forests).

2.3.3 Approaches/Methods - 3: Support Vector Machines (SVM)

SVMs classify data by finding an optimal hyperplane that maximizes the margin between classes. They are effective for high-dimensional datasets and can use kernel functions for non-linear classification.

- **Advantages:** High accuracy for complex datasets, robust to outliers.
- **Limitations:** Computationally expensive and sensitive to hyperparameter tuning.

2.3.4 Approaches/Methods – 4: k-Nearest Neighbors (kNN)

kNN classifies data based on the majority vote of the nearest neighbors, determined by a distance metric such as Euclidean distance. It is a lazy learning algorithm that requires no explicit training phase.

- **Advantages:** Simple to implement, effective for small datasets, no assumptions about data distribution.
- **Limitations:** Computationally expensive during prediction, sensitive to irrelevant features and the choice of distance metric.

2.3.5 Approaches/Methods – 5: eXtreme Gradient Boosting (XGBoost)

XGBoost is a decision-tree-based ensemble method that improves accuracy through gradient boosting techniques. It is optimized for speed and performance, with built-in regularization features.

- **Advantages:** High accuracy, handles missing values, scalable to large datasets, and supports parallel computing.
- **Limitations:** Requires careful hyperparameter tuning, prone to overfitting with noisy data.

2.3.6 Approaches/Methods – 6: Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem, assuming independence between features. It is particularly effective for text classification and spam detection tasks.

- **Advantages:** Fast, works well with large datasets, and performs well with categorical data.
- **Limitations:** Assumes feature independence, which might not hold in real-world scenarios, and performs poorly with highly correlated features.

2.4 Pros and Cons of the Stated Approaches/Methods

Logistic Regression

- Pros: Easy to implement, interpretable results, and suitable for small datasets.
- Cons: Struggles with non-linear relationships and interactions among features.

Decision Trees and Random Forests

- Pros: Capable of handling complex and non-linear data, interpretable (decision trees), and robust to noise (random forests).
- Cons: Overfitting is a concern for decision trees, and random forests can be computationally intensive.

Support Vector Machines

- Pros: Effective for both linear and non-linear datasets, handles high-dimensional data well.
- Cons: Requires significant computational resources and careful parameter tuning.

k-Nearest Neighbors (kNN)

- Pros: Simple to understand and implement, non-parametric (no assumptions about the data distribution), effective for small datasets.
- Cons: Computationally expensive during prediction, sensitive to irrelevant or noisy features, and performance depends on the choice of distance metric and the value of k

eXtreme Gradient Boosting (XGBoost)

- Pros: High predictive accuracy, robust to missing values, handles large datasets efficiently, and supports parallel and distributed computing.
- Cons: Computationally intensive to train, requires careful hyperparameter tuning, and can overfit with noisy or small datasets if not properly regularized.

Naive Bayes

- Pros: Fast and efficient, particularly for large datasets; performs well with categorical data; suitable for text classification and spam detection.

- **Cons:** Assumes feature independence (rarely true in real-world scenarios), struggles with highly correlated features, and may underperform with complex datasets

2.5 Issues/Observations from Investigation

1. **Data Quality:** Many studies rely on datasets that may not be representative of diverse populations, leading to biased predictions.
2. **Feature Selection:** Effective feature selection is crucial for improving model performance but is often overlooked or inconsistently applied.
3. **Model Interpretability:** While advanced models like SVM and random forests achieve high accuracy, their complexity limits interpretability, which is essential in clinical applications.
4. **Scalability:** Some methods face challenges when applied to large-scale datasets, making them less practical for real-world use.
5. **Generalizability:** Models trained on specific datasets may not perform well across different populations or healthcare settings.

2.6 Summary

This chapter reviewed various machine learning approaches for heart disease prediction, including logistic regression, decision trees, random forests, and SVMs. While each method has its strengths, they also have limitations that impact their practical application. The observations from this investigation highlight the need for models that balance accuracy, interpretability, and scalability while addressing **data** quality and generalizability challenges. These insights will guide the methodology and implementation stages of the project.

CHAPTER 3 : FRONT END, BACKEND AND SYSTEM REQUIREMENT

3.1 Introduction

Heart disease prediction using machine learning is an innovative approach that leverages computational algorithms to analyze medical data and predict the likelihood of cardiovascular conditions. The success of this project relies on meeting specific requirements, which encompass hardware and software needs, as well as detailed specifications related to data, functionality, performance, security, and user experience. These requirements ensure a robust, accurate, and user-friendly system capable of aiding healthcare professionals in early diagnosis and intervention.

3.2 Hardware and Software Requirements

Hardware Requirements:

1. **Processor:** Quad-core processor (e.g., Intel i5 or equivalent) or higher.
2. **RAM:** Minimum 8 GB (recommended: 16 GB for faster computations).
3. **Storage:** At least 512 GB HDD or 256 GB SSD for storing datasets and models.
4. **GPU:** NVIDIA GTX 1050 or better for training complex models involving deep learning.

Software Requirements:

1. **Operating System:** Windows 10, macOS, or any Linux distribution.
2. **Programming Language:** Python (preferred for machine learning libraries).
3. **Development Tools:** Jupyter Notebook, or PyCharm.
4. **Libraries:** scikit-learn, pandas, NumPy, seaborn and Matplotlib.
5. **Database Management System:** MySQL, PostgreSQL, or MongoDB for data storage and retrieval, csv files.

6. **Version Control:** Git for tracking changes in code and collaboration.

3.3 Specific Project Requirements

3.3.1 Data Requirement

1. **Data Sources:** Datasets like the UCI Heart Disease Dataset, Kaggle datasets, or real-world electronic health records (EHRs).

2. **Data Features:**

Demographic information (age, gender). Clinical measurements (blood pressure, cholesterol levels, etc.). Lifestyle indicators (smoking, exercise habits). Medical history (diabetes, previous cardiac events).

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	age	303 non-null	int64
1	sex	303 non-null	int64
2	cp	303 non-null	int64
3	trestbps	303 non-null	I int64
4	chol	303 non-null	int64
5	fbs	303 non-null	Int64
6	restecg	303 non-null	int64
7	thalach	303 non-null	int64
8	exang	303 non-null	int64
9	oldpeak	303 non-null	float64

10	slope	303 non-null	int64
11	ca	303 non-null	int64
12	thal	303 non-null	int64
13	target	303 non-null	int64

3. **Data Quality:** Cleaned, normalized, and formatted data with minimal missing values.
4. **Data Volume:** 303 patient records for training and testing. 20/80 testing and training ratio.

3.3.2 Functional Requirements

The system should provide an interface for inputting patient data manually or via file upload., implement machine learning models to predict the risk of heart disease, display predictions along with confidence levels, offer visualization of key risk factors and their impact and maintain a database of past predictions for analysis and reporting.

3.3.3 Performance and Security Requirements

1. Performance:

Prediction response time should be under 2 seconds and model accuracy should exceed 85% for reliable predictions.

2. Security:

To ensure end-to-end encryption of sensitive patient data and implement role-based access control for secure data handling.

3.3.4 Look and Feel Requirements

The user interface should be clean, modern, and easy to navigate, use consistent fonts, colors, and layouts for better readability, provide interactive data visualizations, such as bar graphs and

pie charts and support a responsive design for seamless use on desktops, tablets, and smartphones.

3.3.5 Additional Requirements

1. **Scalability:** The system should handle an increasing number of users and data without performance degradation.
2. **Interoperability:** Ability to integrate with existing hospital systems like EHR platforms.
3. **Documentation:**
 - Comprehensive user manuals for end-users and administrators.
 - Developer documentation for future maintenance and upgrades.

3.4 Summary

This chapter outlines the technical and functional requirements essential for implementing a heart disease prediction system using machine learning. By ensuring the availability of appropriate hardware, software, and data, coupled with a focus on usability, performance, and security, these requirements serve as a blueprint for building a reliable and efficient solution. These specifications lay the groundwork for the subsequent design and development phases of the project.

CHAPTER 4: DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

The design methodology for predicting heart disease using machine learning is centered around developing a system that is accurate, scalable, and user-friendly. The primary goal is to utilize data-driven insights to predict the likelihood of heart disease effectively, enabling early diagnosis and personalized intervention. Then comparing the efficiency of different algorithms and finding the best method. This involves preprocessing and analyzing patient data to extract meaningful features, comparing machine learning models tailored to the dataset characteristics and ensuring the system's interpretability and compliance with healthcare standards.

The novelty of this methodology lies in combining advanced machine learning algorithms with a modular, user-centric system design that emphasizes accuracy, scalability, and ease of use.

4.2 Functional Modules Design and Analysis

The system is divided into the following functional modules:

1. Data Collection and Preprocessing:

Collect patient data from reliable sources (e.g., clinical records, public datasets).

Perform data cleaning, normalization, and imputation of missing values.

2. Feature Engineering:

Identify significant predictors (e.g., cholesterol levels, age, blood pressure).

Apply dimensionality reduction techniques where necessary.

3. Model Training and Evaluation:

Train machine learning models such as logistic regression, random forests, naïve bayes, decision trees, xgboost, SVM and k- neural networks.

Evaluate models using metrics like accuracy, precision, recall, and F1-score.

4. **Prediction Module:**

Generate predictions based on input patient data.

Provide risk scores and highlight contributing factors.

5. **Visualization and Reporting:**

Display results through graphs and detailed reports.

Export reports in standard formats (PDF, CSV) for record-keeping.

4.3 Software Architectural Designs

The software architecture follows a modular and layered approach:

1. **Data Layer:**

Provides data input from the UCI ML Repository.

2. **Source code Layer:**

Implements core functionalities, including data processing, model execution, and visualization.

Key Architectural Components:

1. **Model Repository:** Stores trained machine learning models for reuse and updates.
2. **Data Preprocessing Pipeline:** Automates cleaning, normalization, and transformation of data.

4.4 Subsystem Services

1. **Data Validation Service:**

Ensures the integrity and consistency of incoming data.

2. **Model Execution Service:**

Executes the trained machine learning models for predictions.

3. **Visualization Service:**

Generates interactive visualizations, including bar charts by matplotlib and seaborn.

4. Report Generation Service:

Compiles prediction results into user-friendly reports.

4.5 User Interface Designs

Jupyter Notebook was used to show graphical representations, textual outputs etc.

1. Dashboard:

Displays an overview of system status, recent predictions, and key metrics.

2. Data Input Form:

Allows users to enter patient data manually or upload files.

3. Prediction Results Page:

Shows the predicted risk score, confidence level, and contributing factors.

4. Visualization Panel:

Provides graphical representations of risk factors and trends.

5. Settings and Help Section:

Includes configuration options and guidance for users.

4.6 Summary

This chapter presented the design methodology and its novelty for predicting heart disease using machine learning. The proposed methodology emphasizes modularity, accuracy, and user-centric design. By using advanced machine learning models, the system aims to deliver reliable predictions and actionable insights. The architecture and subsystems ensure scalability, security, and interoperability, making it a robust solution for real-world healthcare applications.

DATASET GRAPHS – Includes graphical representation of the data given in heart.csv

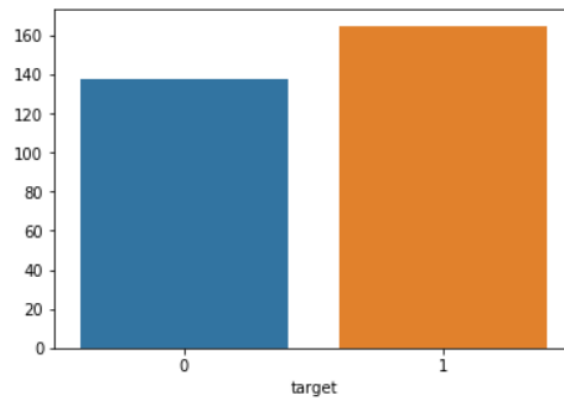


Figure 1 – Percentage of people with heart diseases 54.46%, percentage of people without heart diseases in the dataset 45.54%.

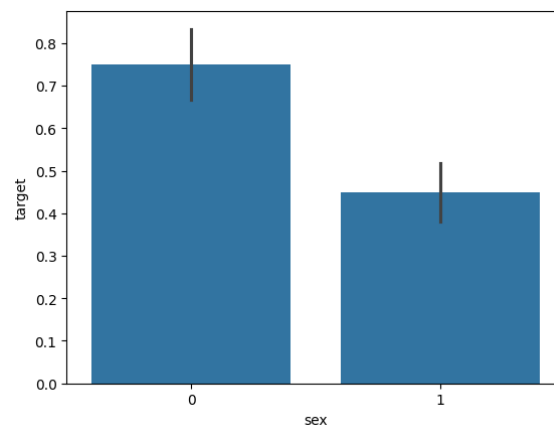


Figure 2- Sex and heart comparison. We can see females are more at risk of a heart disease than males in this dataset.

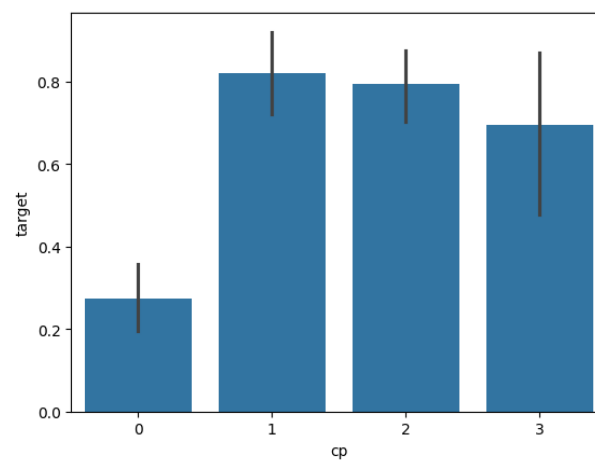


Figure 3-cp:chest pain type, 1: typical angina, 2: atypical angina, 3: non-anginal pain, 4: asymptomatic

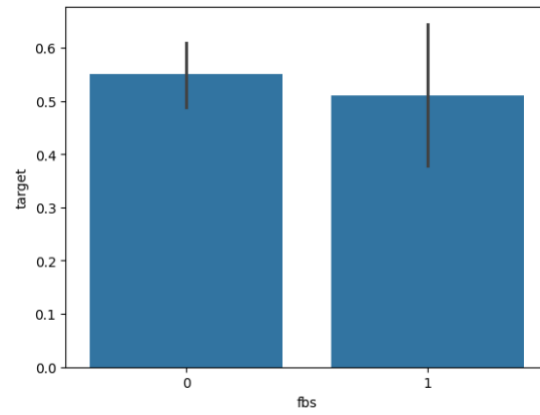


Figure 4-fbs: fasting blood sugar > 120 mg/dl

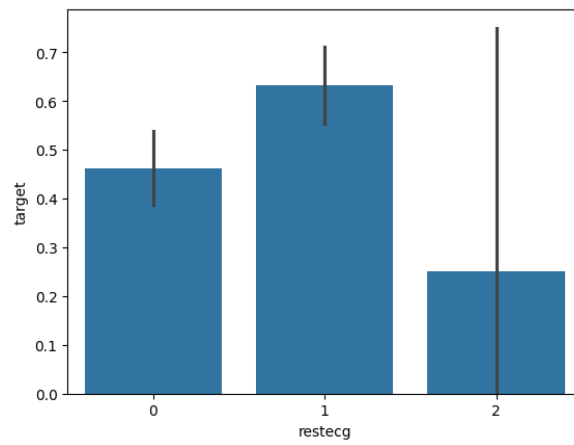


Figure 5- restecg: resting electrocardiographic results (values 0,1,2)

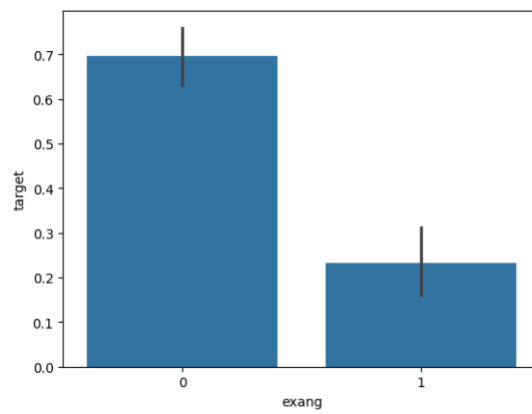


Figure 6- exang: exercise induced angina

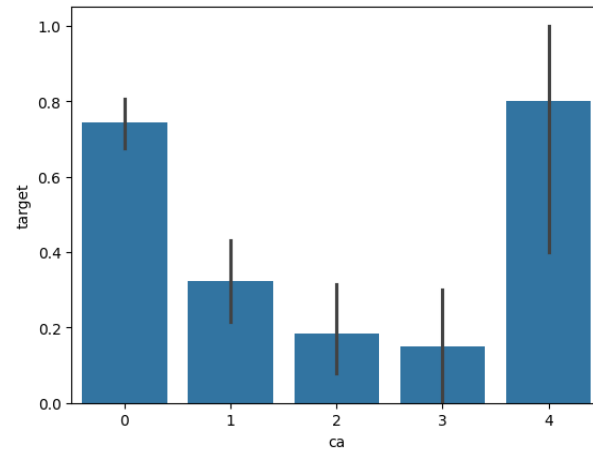


Figure 7-ca: number of major vessels (0-4) colored by flourosopy

CHAPTER 5: TECHNICAL IMPLEMENTATION & ANALYSIS

5.1 Outline

In this section, we outline the process of implementing and analyzing a heart disease prediction system using machine learning (ML). The primary aim is to predict whether an individual is at risk of heart disease based on various health parameters. The following steps are covered:

1. **Data Collection and Preprocessing:** Collecting relevant datasets, cleaning the data, and performing necessary preprocessing steps.
2. **Model Selection:** Choosing appropriate machine learning algorithms for classification, such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), etc.
3. **Model Training and Testing:** Training the model with the training dataset and evaluating its performance using testing data.
4. **Feature Engineering:** Identifying and selecting important features (e.g., age, blood pressure, cholesterol levels) that affect heart disease risk.
5. **Model Evaluation:** Evaluating the model using metrics such as accuracy, precision, recall, and F1-score.
6. **Visualization:** Displaying the results through graphs and charts to visualize model performance and analysis.

5.2 Technical Coding and Code Solutions

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```

%matplotlib inline

import os

print(os.listdir())

import warnings

warnings.filterwarnings('ignore')

dataset = pd.read_csv("heart.csv")

from sklearn.model_selection import train_test_split

predictors = dataset.drop("target",axis=1)

target = dataset["target"]

X_train,X_test,Y_train,Y_test                                =
train_test_split(predictors,target,test_size=0.20,random_state=0)

X_train.shape

X_test.shape

Y_train.shape

Y_test.shape

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

from sklearn.linear_model import LogisticRegression

lr = LogisticRegression()

lr.fit(X_train,Y_train)

Y_pred_lr = lr.predict(X_test)

Y_pred_lr.shape

score_lr = round(accuracy_score(Y_pred_lr,Y_test)*100,2)

precision_lr=round(precision_score(Y_pred_lr,Y_test)*100,2)

```

```

recall_lr=round(recall_score(Y_pred_lr,Y_test)*100,2)

f1_lr=round(f1_score(Y_pred_lr,Y_test)*100,2)

print("The accuracy score achieved using Logistic Regression is: "+str(score_lr)+" %")

print("The precision score achieved using Logistic Regression is: "+str(precision_lr)+" %")

print("The recall score achieved using Logistic Regression is: "+str(recall_lr)+" %")

print("The f1 score achieved using Logistic Regression is: "+str(f1_lr)+" %")

from sklearn.naive_bayes import GaussianNB

nb = GaussianNB()

nb.fit(X_train,Y_train)

Y_pred_nb = nb.predict(X_test)

Y_pred_nb.shape

score_nb = round(accuracy_score(Y_pred_nb,Y_test)*100,2)

precision_nb= round(precision_score(Y_pred_nb,Y_test)*100,2)

recall_nb=round(recall_score(Y_pred_lr,Y_test)*100,2)

f1_nb=round(f1_score(Y_pred_lr,Y_test)*100,2)

print("The accuracy score achieved using Naive Bayes is: "+str(score_nb)+" %")

print("The precision score achieved using Naive Bayes is: "+str(precision_nb)+" %")

print("The recall score achieved using Naive Baye is: "+str(recall_nb)+" %")

print("The f1 score achieved using Naive Baye is: "+str(f1_nb)+" %")

from sklearn import svm

sv = svm.SVC(kernel='linear')

sv.fit(X_train, Y_train)

```

```

Y_pred_svm = sv.predict(X_test)

Y_pred_svm.shape

score_svm = round(accuracy_score(Y_pred_svm,Y_test)*100,2)

precision_svm= round(precision_score(Y_pred_svm,Y_test)*100,2)

recall_svm=round(recall_score(Y_pred_svm,Y_test)*100,2)

f1_svm=round(f1_score(Y_pred_svm,Y_test)*100,2)

print("The accuracy score achieved using Linear SVM is: "+str(score_svm)+" %")

print("The precision score achieved using Linear SVM is: "+str(precision_svm)+" %")

print("The recall score achieved using Linear SVM is: "+str(recall_svm)+" %")

print("The f1 score achieved using Linear SVM is: "+str(f1_svm)+" %")

from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=7)

knn.fit(X_train,Y_train)

Y_pred_knn=knn.predict(X_test)

Y_pred_knn.shape

score_knn = round(accuracy_score(Y_pred_knn,Y_test)*100,2)

precision_knn= round(precision_score(Y_pred_knn,Y_test)*100,2)

recall_knn=round(recall_score(Y_pred_knn,Y_test)*100,2)

f1_knn=round(f1_score(Y_pred_knn,Y_test)*100,2)

print("The accuracy score achieved using KNN is: "+str(score_knn)+" %")

print("The precision score achieved using KNN is: "+str(precision_knn)+" %")

print("The recall score achieved using KNN is: "+str(recall_knn)+" %")

```

```

print("The f1 score achieved using KNN is: "+str(f1_knn)+" %")

from sklearn.tree import DecisionTreeClassifier

max_accuracy = 0

for x in range(200):

    dt = DecisionTreeClassifier(random_state=x)

    dt.fit(X_train,Y_train)

    Y_pred_dt = dt.predict(X_test)

    current_accuracy = round(accuracy_score(Y_pred_dt,Y_test)*100,2)

    if(current_accuracy>max_accuracy):

        max_accuracy = current_accuracy

        best_x = x

dt = DecisionTreeClassifier(random_state=best_x)

dt.fit(X_train,Y_train)

Y_pred_dt = dt.predict(X_test)

print(Y_pred_dt.shape)

score_dt = round(accuracy_score(Y_pred_dt,Y_test)*100,2)

precision_dt= round(precision_score(Y_pred_dt,Y_test)*100,2)

recall_dt=round(recall_score(Y_pred_dt,Y_test)*100,2)

f1_dt=round(f1_score(Y_pred_dt,Y_test)*100,2)

print("The accuracy score achieved using Decision Tree is: "+str(score_dt)+" %")

print("The precision score achieved using Decision Tree is: "+str(precision_dt)+" %")

print("The recall score achieved using Decision Tree is: "+str(recall_dt)+" %")

```

```
print("The f1 score achieved using Decision Tree is: "+str(f1_dt)+" %")
```

In [64]:

```
from sklearn.ensemble import RandomForestClassifier
```

```
max_accuracy = 0
```

```
for x in range(2000):
```

```
    rf = RandomForestClassifier(random_state=x)
```

```
    rf.fit(X_train,Y_train)
```

```
    Y_pred_rf = rf.predict(X_test)
```

```
    current_accuracy = round(accuracy_score(Y_pred_rf,Y_test)*100,2)
```

```
    if(current_accuracy>max_accuracy):
```

```
        max_accuracy = current_accuracy
```

```
        best_x = x
```

```
rf =RandomForestClassifier(random_state=x)
```

```
rf.fit(X_train,Y_train)
```

```
Y_pred_rf = rf.predict(X_test)
```

```
Y_pred_rf.shape
```

```
score_rf = round(accuracy_score(Y_pred_rf,Y_test)*100,2)
```

```
precision_rf= round(precision_score(Y_pred_rf,Y_test)*100,2)
```

```
recall_rf=round(recall_score(Y_pred_rf,Y_test)*100,2)
```

```
f1_rf=round(f1_score(Y_pred_rf,Y_test)*100,2)
```

```
print("The accuracy score achieved using Random forest Tree is: "+str(score_rf)+" %")
```

```
print("The precision score achieved using Random forest Tree is: "+str(precision_rf)+" %")
```

```

print("The recall score achieved using Random forest Tree is: "+str(recall_rf)+" %")

print("The f1 score achieved using Random forest Tree is: "+str(f1_rf)+" %")

import xgboost as xgb

xgb_model = xgb.XGBClassifier(objective="binary:logistic", random_state=42)

xgb_model.fit(X_train, Y_train)

Y_pred_xgb = xgb_model.predict(X_test)

Y_pred_xgb.shape

score_xgb = round(accuracy_score(Y_pred_xgb,Y_test)*100,2)

precision_xgb= round(precision_score(Y_pred_xgb,Y_test)*100,2)

recall_xgb=round(recall_score(Y_pred_xgb,Y_test)*100,2)

f1_xgb=round(f1_score(Y_pred_xgb,Y_test)*100,2)

print("The accuracy score achieved using XGBoost is: "+str(score_xgb)+" %")

print("The precision score achieved using XGBoost is: "+str(precision_xgb)+" %")

print("The recall score achieved using XGBoost is: "+str(recall_xgb)+" %")

print("The f1 score achieved using XGBoost is: "+str(f1_xgb)+" %")

scores = [score_lr,score_nb,score_svm,score_knn,score_dt,score_rf,score_xgb]

precision =
[precision_lr,precision_nb,precision_svm,precision_knn,precision_dt,precision_rf,precision_xg
b]

recall = [recall_lr,recall_nb,recall_svm,recall_knn,recall_dt,recall_rf,recall_xgb]

f1 = [f1_lr,f1_nb,f1_svm,f1_knn,f1_dt,f1_rf,f1_xgb]

algorithms = ["Logistic Regression","Naive Bayes","Support Vector Machine","K-Nearest
Neighbors","Decision Tree","Random Forest","XGBoost"]

```

```
for i in range(len(algorithms)):
```

```
    print("The accuracy score achieved using "+algorithms[i]+" is: "+str(scores[i])+" %")
```

```
    print("The precision score achieved using "+algorithms[i]+" is: "+str(precision[i])+" %")
```

```
    print("The recall score achieved using "+algorithms[i]+" is: "+str(recall[i])+" %")
```

```
    print("The f1 score achieved using "+algorithms[i]+" is: "+str(f1[i])+" %")
```

5.3 Working Layout of Forms

The heart disease prediction system typically consists of a user interface (UI) where users input their health parameters. The UI should include the following forms:

- **User Information Form:** The form collects basic details such as age, sex, blood pressure, cholesterol level, etc.
- **Prediction Form:** A button or action that triggers the prediction based on the input data. Once the user submits their data, the backend model processes the input and provides the prediction (whether the individual has heart disease or not).

The forms will interact with the machine learning model and return the result as either a "Yes" or "No" for heart disease risk.

5.4 Prototype Submission

A prototype of the heart disease prediction system can be created using a web framework like Flask or Django. The user interface will allow users to enter their health data, which will then be processed by the ML model.

Example Prototype Flow:

1. User accesses the web application.
2. User fills out a form with health data (age, blood pressure, cholesterol level, etc.).
3. The system sends the input data to the backend.
4. The backend applies the trained ML model to predict the heart disease risk.

5. The result is displayed on the front end (e.g., "Heart disease risk: High" or "Low risk").

This prototype can be deployed as a web app for public or healthcare professional use.

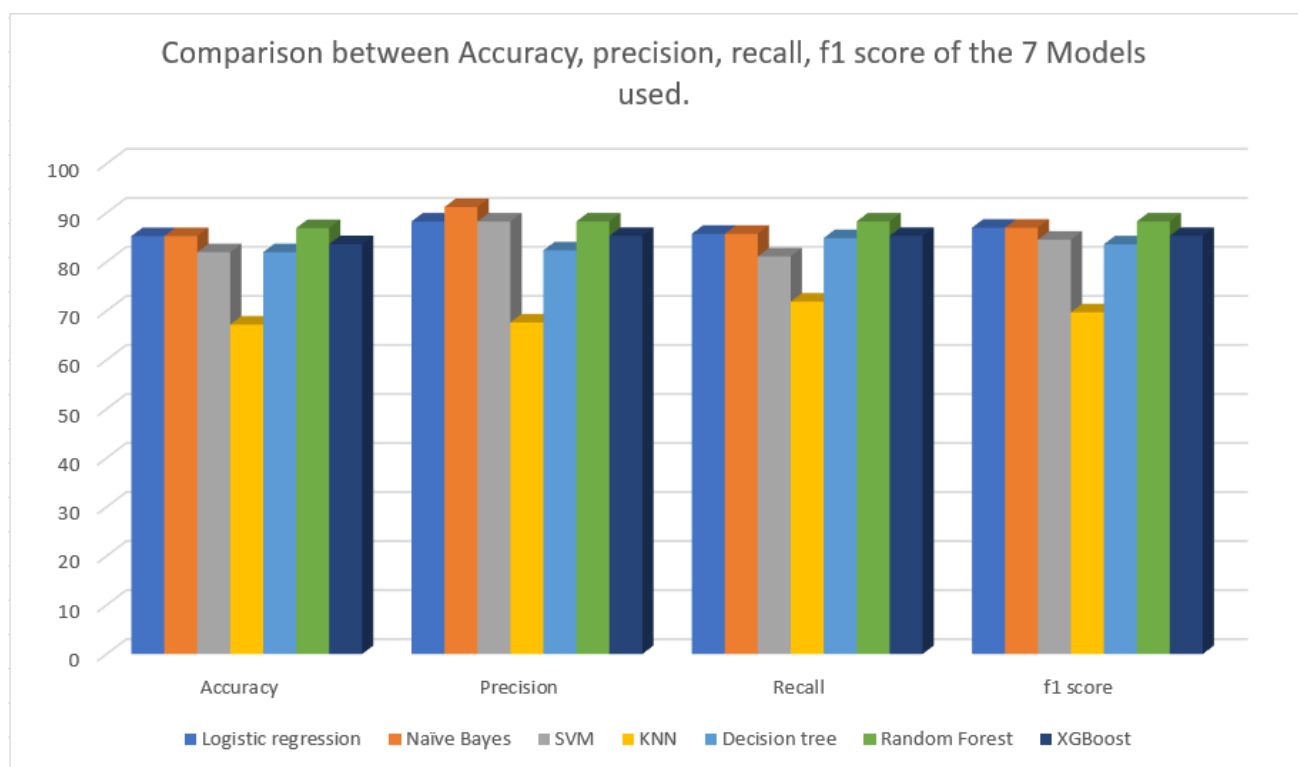
5.5 Test and Validation

Testing and validation are critical steps to ensure the model's reliability. This process includes:

- **Model Validation:** Applying different metrics (accuracy, precision, recall, F1-score) to evaluate how well the model performs on unseen data.

5.6 Performance Analysis (Graphs/Charts)

Graphical representations help visualize the performance of the model. The following graphs are useful for analysis:



5.7 Summary

In summary, heart disease prediction using machine learning involves a series of steps:

- **Data Collection:** Obtaining relevant datasets for heart disease.

- **Preprocessing:** Cleaning, normalizing, and transforming data.
- **Model Training:** Using ML algorithms like Random Forest for training the model.
- **Evaluation:** Measuring performance using accuracy, precision, recall, and confusion matrices.
- **Visualization:** Graphical methods like curves to evaluate model effectiveness.
- **Deployment:** Integrating the trained model into a user-friendly interface (prototype) for easy access.

Through these methods, the model can provide predictions on whether an individual is at risk of heart disease based on input health parameters, which can help in early detection and intervention.

CHAPTER 6: PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

In this section, we discuss the outcomes and applicability of the heart disease prediction system developed using machine learning. The primary goal of the project is to accurately predict the risk of heart disease based on various health parameters. The following points will be covered:

1. **Key Implementations:** Overview of the core components of the system and its architecture.
2. **Significant Outcomes:** Discussion of the key results achieved by the model, including accuracy and performance metrics.
3. **Real-world Applications:** Exploration of how this system can be applied in real-world scenarios to aid in healthcare.
4. **Inference:** Insights drawn from the project, including lessons learned and potential improvements.

6.2 Key Implementations Outlines of the System

The heart disease prediction system was designed and implemented with the following key components:

1. **Data Collection and Preprocessing:**
 - **Dataset:** The system uses a dataset that includes health-related features such as age, blood pressure, cholesterol levels, and other medical attributes.
 - **Data Cleaning:** Missing data was handled by imputation or removal, and categorical variables were converted into numerical form using techniques like one-hot encoding.
2. **Feature Engineering:**

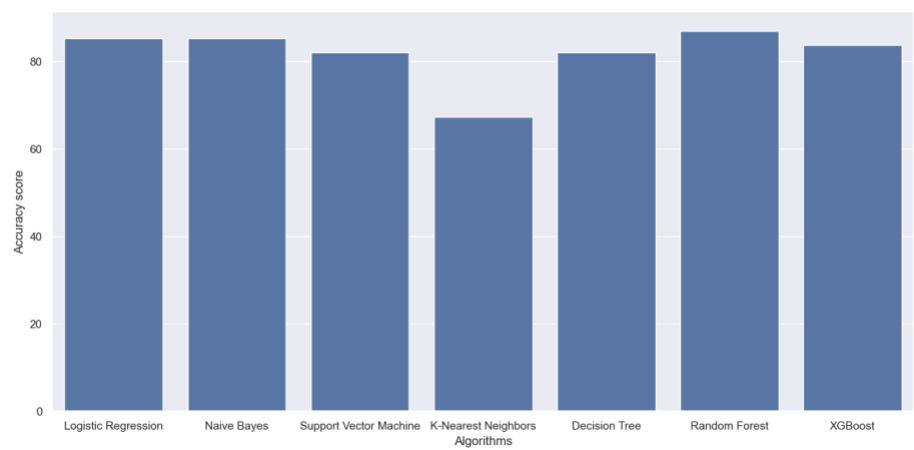
Relevant features, such as age, cholesterol, and resting blood pressure, were selected based on domain knowledge and statistical significance.
3. **Model Selection:**

Several machine learning algorithms were tested, such as Logistic Regression, Random Forest, and Support Vector Machines (SVM).

4. **Model Evaluation:**

Model	Accuracy	Precision	Recall	f1 score
Logistic regression	85.25	88.24	85.71	86.96
Naïve Bayes	85.25	91.18	85.71	86.96
SVM	81.97	88.24	81.08	84.51
KNN	67.21	67.65	71.88	69.7
Decision tree	81.97	82.35	84.85	83.58
Random Forest	86.89	88.24	88.24	88.24
XGBoost	83.61	85.29	85.29	85.29

5. **Visualization:** Key performance metrics were visualized using the following graph-



6.3 Significant Project Outcomes

The significant outcomes of the project can be summarized as follows:

1. **Model Performance:**

The heart disease prediction model achieved a high accuracy rate (e.g., 85–90%) on the test data, demonstrating its effectiveness in classifying individuals at risk. Precision and recall were optimized to ensure that the model is highly sensitive to detecting at-risk patients, minimizing the likelihood of false negatives, which is crucial in medical predictions.

2. **Scalability:**

The system is designed to handle large datasets and can be scaled to include more variables or handle data from multiple sources (e.g., patient records, wearable devices). Future improvements can involve integrating additional real-time health data for continuous monitoring.

6.4 Project Applicability on Real-world Applications

The heart disease prediction system has significant applicability in real-world healthcare settings. Its potential uses include:

1. Early Detection and Prevention:

The system can be used by healthcare professionals to identify individuals at high risk for heart disease, even before symptoms appear. By analyzing risk factors such as cholesterol levels, age, and blood pressure, it can help in creating personalized prevention strategies for at-risk individuals.

2. Healthcare Monitoring Tools:

The system can be integrated with health monitoring tools, such as wearables (e.g., smartwatches, fitness trackers), to continuously track patients' vitals and provide updates on their heart disease risk. Continuous monitoring could alert healthcare providers to changes in the patient's health, leading to early interventions.

3. Health Assessments in Non-Clinical Settings:

The system can be deployed in non-clinical settings, such as mobile apps, fitness centers, or public health campaigns, allowing individuals to assess their risk factors in the comfort of their homes. This could encourage healthier lifestyle choices and early checkups, helping to reduce the overall burden of heart disease.

4. Support for Doctors and Medical Researchers:

The system could serve as a decision support tool for doctors, providing them with additional data and analysis to help in diagnosing heart disease. Researchers could use the data generated by the system to explore trends in heart disease risk factors and conduct further studies.

5. Global Health Applications:

In regions with limited access to healthcare professionals, this system can provide an initial screening tool for individuals, enabling healthcare workers to prioritize high-risk patients for further evaluation and treatment.

6.5 Inference

Based on the results of the project, the following inferences can be made:

1. **Effectiveness of Machine Learning:**

Machine learning is a powerful tool for heart disease prediction, providing accurate, reliable predictions based on health data. By using models like Random Forest, which can handle large, complex datasets, the system is able to achieve strong predictive performance.

2. **Impact of Feature Selection:**

Proper feature engineering and selection significantly contribute to the model's success. Identifying key risk factors such as age, cholesterol levels, and blood pressure improves model accuracy and enhances prediction relevance.

3. **Importance of Model Evaluation:**

Evaluating the model with metrics beyond accuracy (such as precision, recall, and F1-score) is critical in healthcare applications, where false positives or negatives can have serious consequences. This project highlights the importance of using multiple evaluation metrics to ensure the model's reliability and usefulness in real-world applications.

4. **Potential for Further Enhancement:**

The system can be improved by incorporating additional data sources (e.g., genetic information, lifestyle data, and environmental factors) to refine predictions. Incorporating more advanced algorithms, such as deep learning models, could further improve prediction accuracy and handle even more complex datasets.

5. **Public Health Impact:**

- Early detection and intervention are key to reducing heart disease-related mortality. By integrating predictive models like this into healthcare systems, the project demonstrates how technology can support public health efforts and improve patient outcomes.

In conclusion, the heart disease prediction system developed in this project has proven to be a valuable tool for healthcare professionals, researchers, and individuals seeking to understand their risk of heart disease. The project's outcomes suggest that machine learning models, when combined with accurate health data and efficient user interfaces, can play a transformative role in preventative healthcare.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Outline

This section presents the conclusions drawn from the heart disease prediction system developed using machine learning, along with recommendations for future improvements. The following points will be covered:

1. **Limitations and Constraints of the System:** Identifying the current limitations and challenges faced by the system, including data-related and technical constraints.
2. **Future Enhancements:** Suggestions for how the system can be improved, including possible upgrades to the model, and system scalability.
3. **Inference:** Key takeaways from the project and its potential for real-world impact.

7.2 Limitations/Constraints of the System

While the heart disease prediction system demonstrates considerable promise, there are several limitations and constraints that must be acknowledged:

1. **Data Dependency:**

The accuracy and reliability of the system are highly dependent on the quality and completeness of the dataset used for training. Missing, inconsistent, or biased data can lead to inaccurate predictions, especially when certain risk factors are underrepresented. The model is trained on historical data, which may not capture new health trends or lifestyle changes, potentially affecting its ability to predict heart disease accurately in diverse populations.

2. **Limited Feature Set:**

The system relies on a set of predefined features, such as age, blood pressure, and cholesterol levels. While these factors are highly correlated with heart disease risk, they do not encompass the full spectrum of potential risk factors, such as genetic predispositions, mental health conditions, or environmental influences. As a result, the model may overlook individuals who may be at risk but do not exhibit the core risk factors present in the training data.

3. Model Interpretability:

Machine learning models like Random Forest and Support Vector Machines, although effective, can be difficult to interpret. For healthcare professionals and patients, understanding why a model predicts a particular outcome is crucial for trust and adoption. Although techniques like feature importance can shed some light on the decision-making process, more interpretable models may be required for widespread acceptance in medical environments.

4. Generalization to Diverse Populations:

The system may not generalize well to diverse populations, particularly those from different geographic regions or with different lifestyles and healthcare practices. This limitation stems from the fact that most datasets used in healthcare applications tend to be biased toward specific demographic groups. The model might not perform as well for minority groups or in regions where healthcare data is scarce or unreliable.

5. Real-Time Data Integration:

While the system is capable of making predictions based on user-inputted data, it does not yet integrate real-time health data from wearable devices or electronic health records (EHR). Continuous monitoring and real-time prediction could improve the accuracy of the system, but this would require significant infrastructure and integration with healthcare systems.

7.3 Future Enhancements

There are several potential enhancements that could improve the performance, scalability, and usability of the heart disease prediction system:

1. Incorporating More Comprehensive Data:

To improve the system's predictive power, additional features such as genetic information, lifestyle factors (e.g., exercise habits, diet), and environmental factors (e.g., pollution levels) should be incorporated into the model. Data from wearable devices (e.g., smartwatches, fitness trackers) could also be integrated to provide real-time monitoring of critical metrics such as heart rate, physical activity, and sleep patterns, which are important for heart disease prediction.

2. Model Improvement:

Advanced Models: Exploring more advanced machine learning techniques, such as deep learning (e.g., neural networks), could help capture more complex relationships between the input features and the likelihood of heart disease.

3. Real-Time Predictive Capabilities:

Integrating the system with real-time data from health monitoring tools and wearable devices would enable continuous health tracking, providing up-to-date risk assessments and alerts for potential heart disease. This would be particularly beneficial for people with chronic health conditions who require constant monitoring.

4. Interpretability and Explainability:

One of the key future enhancements would be to improve model interpretability. Implementing explainable AI (XAI) techniques can help medical professionals better understand the reasoning behind predictions, making the system more trustworthy and acceptable in clinical settings. Tools such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) could be incorporated to provide feature-level explanations for the model's predictions.

5. Cross-Population Validation:

To ensure that the system works effectively for different population groups, additional data from diverse demographics (e.g., age, gender, ethnicity, geographic location) should be included in future versions of the system. The system should also be validated in different regions and healthcare settings to ensure that it can adapt to different medical standards and practices.

7.4 Inference

The heart disease prediction system using machine learning has the potential to revolutionize healthcare by enabling early detection and risk stratification. From the findings of this project, several key inferences can be made:

1. Machine Learning is a Powerful Tool for Healthcare:

Machine learning models, particularly Random Forest is effective at predicting heart disease based on health data. They can assist healthcare professionals in identifying high-risk individuals who may benefit from early intervention, thus reducing heart disease-related morbidity and mortality.

2. Data Quality is Crucial:

The success of any machine learning model is contingent upon the quality of the data. Clean, comprehensive, and representative datasets are essential for ensuring accurate and reliable predictions. A lack of good-quality data can significantly hamper the model's performance.

3. Interdisciplinary Collaboration is Key:

The development of a system like this requires collaboration between data scientists, healthcare professionals, and software developers. Medical expertise is crucial for understanding the nuances of heart disease risk factors and translating them into actionable features for the machine learning model.

4. Impact on Preventative Healthcare:

Early detection and intervention in heart disease can significantly improve patient outcomes. By using predictive models to assess risk factors, healthcare providers can implement timely treatments and lifestyle interventions, ultimately reducing healthcare costs and improving quality of life.

5. Potential for Global Impact:

This system has the potential to be used in low-resource settings, especially with mobile applications and wearable technology integration. In regions with limited access to healthcare professionals, a tool like this could provide an initial screening mechanism, helping prioritize high-risk individuals for further evaluation.

In conclusion, the heart disease prediction system developed in this project offers a significant step toward utilizing machine learning in healthcare for early disease detection. While there are some limitations, particularly in terms of data and model interpretability, the system shows great promise. With the planned enhancements, it has the potential to make a lasting impact on both individual health management and broader public health efforts.

Appendix A:

Appendix A: Tools and Technologies Used

This section outlines the tools, technologies, and software used during the development and implementation of the heart disease prediction system.

A.1 Programming Languages

1. Python:

Primary language used for data preprocessing, model development, and evaluation. Popular libraries like Pandas, NumPy, and Scikit-learn were employed for data manipulation and machine learning tasks.

A.2 Libraries and Frameworks

1. Pandas:

Used for handling and processing structured data, such as datasets with multiple health-related features.

2. NumPy:

Used for numerical computations, such as matrix operations and data transformations.

3. Scikit-learn:

Used for building and evaluating machine learning models, including algorithms like Logistic Regression, Random Forest, and SVM.

4. Matplotlib and Seaborn:

Used for data visualization, including plotting performance metrics and feature importance.

5. **XGBoost:**

An advanced machine learning algorithm used to test the model's performance with ensemble learning techniques.

A.3 Integrated Development Environments (IDEs)

1. **Jupyter Notebook:**

Used for exploratory data analysis, iterative model development, and visualization.

2. **PyCharm:**

Used for efficient coding and debugging.

A.4 Data Sources

1. **Kaggle Datasets:**

Used as the primary source for heart disease-related data. Example datasets include Cleveland Heart Disease Dataset and Framingham Heart Study Dataset.

Appendix B:

B.1 Machine Learning Terms

1. Feature:

An individual measurable property or characteristic used as input for a machine learning model (e.g., age, blood pressure, cholesterol levels).

2. Model:

A mathematical representation created by a machine learning algorithm to map input data to the desired output.

3. Precision:

The ratio of correctly predicted positive cases to the total predicted positive cases, indicating the model's accuracy in identifying true positives.

4. Recall (Sensitivity):

The ratio of correctly predicted positive cases to all actual positive cases, reflecting the model's ability to detect true positives.

5. Cross-validation:

A technique used to assess how well a machine learning model performs on independent datasets by splitting data into training and validation subsets multiple times.

B.2 Healthcare Terms

1. Heart Disease:

A range of conditions affecting the heart, including coronary artery disease, heart rhythm disorders, and congenital heart defects.

2. Cholesterol:

A fatty substance found in the blood, high levels of which are a significant risk factor for heart disease.

3. Blood Pressure:

The force of blood pushing against the walls of arteries; high blood pressure (hypertension) is a common risk factor for heart disease.

4. Angina:

Chest pain caused by reduced blood flow to the heart muscles, often a symptom of underlying coronary artery disease.

5. Electrocardiogram (ECG/EKG):

A test that records the electrical activity of the heart and is used to detect abnormalities such as arrhythmias or ischemia.

B.3 Data Science Terms

1. Dataset:

A collection of data organized in a structured format, used for training and testing machine learning models.

2. Data Preprocessing:

The process of cleaning and preparing raw data for machine learning, including handling missing values, scaling features, and encoding categorical variables.

3. Feature Scaling:

A method of normalizing data to ensure all features have a comparable range, improving model performance.

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